THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE UNIT TEST SET 2

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A change in which one or more new substances are formed is a: 1

(a) reversible change only **(b)** chemical change
(c) change of state **(d)** physical change

2. The formula of sulphuric acid is: 1

(a) $NaOH$ **(b)** H_2SO_4 **(c)** HCl **(d)** HNO_3

3. Materials that do not allow electric current to pass through them are: 1

(a) conductors **(b)** filaments **(c)** insulators **(d)** metals

4. Which of the following is a non-metal in the gaseous state? 1

(a) mercury **(b)** sodium **(c)** oxygen **(d)** zinc

5. Which of the following is an irreversible change? 1

(a) ripening of fruit **(b)** folding paper
(c) melting ice **(d)** freezing water

6. Neutral substances: 1

(a) turn blue litmus red **(b)** do not change the colour of litmus
(c) turn red litmus blue **(d)** turn both red

7. The heating effect of electric current is used in: 1

(a) radio **(b)** electric iron **(c)** compass **(d)** fan only

8. Metal oxides are generally: 1

(a) basic **(b)** neutral **(c)** acidic **(d)** sour

9. Rust is chemically: 1

(a) iron sulphide **(b)** hydrated iron oxide
(c) iron carbonate **(d)** iron chloride

10. Which of these is acidic? 1

(a) baking soda solution **(b)** soap solution
(c) lemon juice **(d)** lime water

11. The path along which electric current flows is called an: 1

(a) electric switch **(b)** electric circuit
(c) electric wire only **(d)** electric field

12. The property by which metals can be drawn into wires is: 1

(a) ductility **(b)** hardness **(c)** lustre **(d)** malleability

13. Melting of ice is a: 1

(a) decomposition **(b)** physical change **(c)** burning **(d)** chemical change

14. The formula of sodium hydroxide is: 1

(a) KOH **(b)** $NaOH$ **(c)** $Ca(OH)_2$ **(d)** HCl

15. Which of these is a good conductor of electricity? 1

(a) copper **(b)** wood **(c)** plastic **(d)** rubber

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. How can you make a battery from cells? 2

17. Name a metal and a non-metal that are liquid at room temperature. 2

18. What are the conditions for rusting? 2

19. Write the formulae of hydrochloric acid and sodium hydroxide. 2

20. Draw the symbols (describe) for cell and bulb. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Describe an activity to show a chemical change using vinegar and baking soda. 3

22. Why does a tooth decay get worse with acid? 3

23. What happens when two cells are connected in series with the same polarity? 3

24. Describe the reaction of metals with acids. 3

OR

How can one identify a chemical change by a change in smell or taste? Give an example. 3

25. Make a table of the colours of three indicators with acid and base. 3

OR

Explain why a bulb does not glow when a rubber piece is connected between two wires. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain acids found in food items and the natural source of each. 5

OR

What is a battery? Explain how three 1.5 V cells in series produce 4.5 V and what happens to bulb brightness. 5

27. Describe exceptions in properties of metals and non-metals with examples. 5

OR

What are the indications of chemical change? Describe three activities. 5

END OF QUESTION PAPER — Total: 50 Marks

INSTAMIND